

Social Hardware

Using the power of generative design to create new-age prosthetics. We speak to Cameron Norris and Abhit K., Co-Founders of Social Hardware.

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We were introduced to Social Hardware by Autodesk at Hyderabad Design Week. Social Hardware works with amputee rehabilitation centres to help amputees get the prosthetics they need on the basis of their individual requirements. One of the key components of their product is the Avocado Wrist Connector which is compatible with popular prosthetic designs and was designed using Autodesk's generative design tools.

"We're very thrilled to have one of our new-age innovators, Social Hardware launch their product at the Autodesk Design Night," said Priya Balijepalli, Sustainability Success Manager, Autodesk India. "Generative design is a design exploration process. It's empowered by human-machine collaboration. We are stretching the capabilities of human creativity, empowered by Machine Learning, to give the designer thousands of options for the product that they want to design."

d Assume that the user is indulging in a certain activity, say ploughing, there will be stress forces acting along a certain edge and segment of the Avocado Wrist Connector. For a different task, the stress forces might be acting along a different edge and segment. So do you create different variants to cater to different use cases?

Cameron Norris: We made the Connector compatible with essentially any activity within a reasonable range of activities. The maximum stress is 50 kg whatever the activity might be and it'll be able to resist that.

Abhit K.: That is part of the design simulation process where we are able to

understand the forces and the points of actions. For example, if the person is using a mechanical hammer at a construction site then we look at the force that will be acting at the point where the tool is connected and whether it will break or not. At the same time, the material that we're using for the tool is stainless steel. If you were using wood or plastic, the stress level would have been much higher. The tool has such a good fit that it holds well and can handle 50 kgs of force which is what would be experienced at construction sites.

d Generative design is more of a constraint-based approach where you provide constraints and the rest of the design process is handled by the AI. While designing the Avocado Wrist Connector, did you provide Steel as a constraint or was it something that the AI suggested as part of the iterative process?



Social Hardware's Avocado Wrist Connector

Cameron Norris: We tested various materials and made a decision ourselves. So rather than the AI dictating what we do, we take the output that's generated through the software, have a look at what's most suitable for manufacturing and then we proceed. It's more of a collaboration between the AI and the human designers which lets us reach the optimum solution. In the case of stainless steel, it's mass-manufacturable, so we can use CNC milling which is relatively low-cost, and worked towards what we set out to achieve.

Abhit K.: I think generative design allows material selection so you can test various materials and see different views on the basis of the chosen material. Then we perform further simulations to see which works out better. It has definitely provided us a platform to see the behaviour of different materials.

Cameron Norris: It's kind of a compromise between manufacturability, durability, weight and cost.



Prototype prosthetic with the Avocado Wrist Connector installed

d Going by what is usually showcased as examples of generative design, we usually see designs that are often asymmetric and organic. The Avocado Connector seems like a very traditional design philosophy. Where's the generative design element here?

Abhit K.: When we first designed the product, the weight was 300 grams and when we used generative design, the result was weighing 56 grams. If you observe that output, it's a very organic and alien-looking design. But the fact is that we have to take into account that the design has to be manufacturable and needs to have a product fit. That's where the human-AI collaboration comes in. The way we see it, generative design reduces the complexity around the initial design and provides us with an alternative, onto which, we have to perform an improvement in terms of aesthetics, making it market-friendly and manufacturable.

Cameron Norris: Rather than using the generative output directly, we use it as a guideline. So that shows us where we can reduce materials to achieve the initial goal that we set out to do. It's essentially a template. **d**